Introductory Econometrics

The Simple Regression Model

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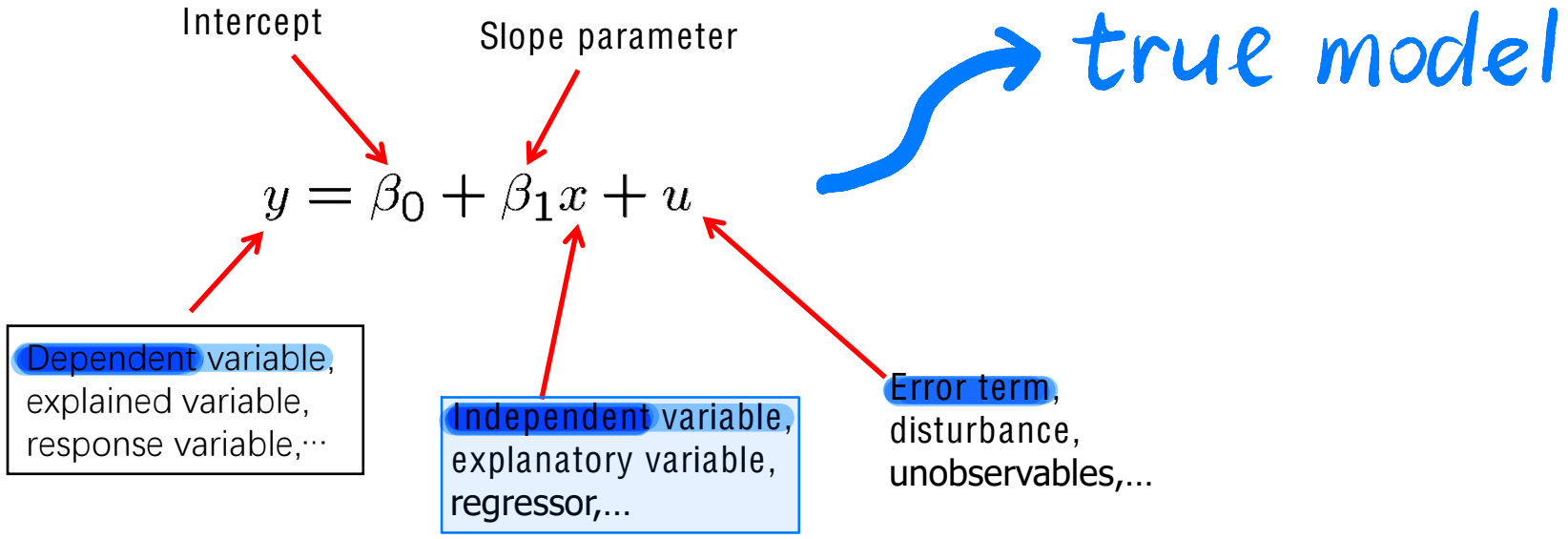
Outline

- Definition of the Simple Regression Model
- Deriving the Ordinary Least Squares Estimates
- Properties of OLS on any Sample of Data
- Units of Measurement and Functional Form
- Testing
- Confidence interval

Definition of the simple linear regression model

- Definition of the simple linear regression model
- For instance, is amount of fertilizer, and is soybean yield; is years of schooling, is hourly wage.

"Explains variable y in terms of variable x"



Definition of the simple linear regression model

- Interpretation of the simple linear regression model
- All other factors that affect are in . We want to know how y changes when x changes, holding u fixed.

"Studies how y varies with changes in x:"

"ceteris paribus"

$$\frac{\Delta y}{\Delta x} = \beta_1 \quad \text{as long as} \quad \frac{\Delta u}{\Delta x} = 0$$

By how much does the dependent variable change if the independent variable is increased by one unit?

Interpretation only correct if all other things remain equal when the independent variable is increased by one unit

- The simple linear regression model is rarely applicable in practice but its discussion is useful for pedagogical reasons

Definition of the simple linear regression model

- Example: Soybean yield and fertilizer

$$yield = \beta_0 + \beta_1 fertilizer + u$$

Rainfall,
land quality,
presence of parasites, ...

Measures the effect of fertilizer on yield, holding all other factors fixed

- Example: A simple wage equation

$$wage = \beta_0 + \beta_1 educ + u$$

Labor force experience,
tenure with current employer,
work ethic, intelligence, ...

Measures the change in hourly wage given another year of education, holding all other factors fixed

When is there a causal interpretation?

- **When is there a causal interpretation?**

➢How can we hope to learn in genral about the ceteris paribus effect of x on y, holding other factors fixed, when we are ignoring all those other factors?

➢What if x and u are related?

- **Example: wage equation**

$$wage = \beta_0 + \beta_1 educ + u$$ ← e.g. intelligence …

The conditional mean independence assumption is unlikely to hold because individuals with more education will also be more intelligent on average.

When is there a causal interpretation?

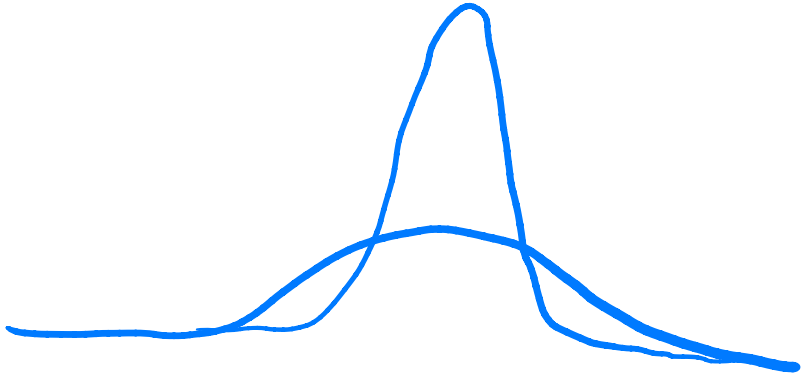
- **Zero conditional mean independence assumption** (a weaker assumption)

  1. Normalizing u

$$E(u) = 0$$

  2. u is mean independent of x

$$E(u|x) = E(u)$$



截图(Alt + A)

the average value if the unobservable is the same across all slices of the population determined by the value of x .

Combined:

$$E(u|x) = 0$$

The explanatory variable must not contain information about the mean of the unobserved factors

- **Example: education and "ability"**

Population regression function (PFR)

- Population regression function (PFR)
  - The conditional mean independence assumption implies that

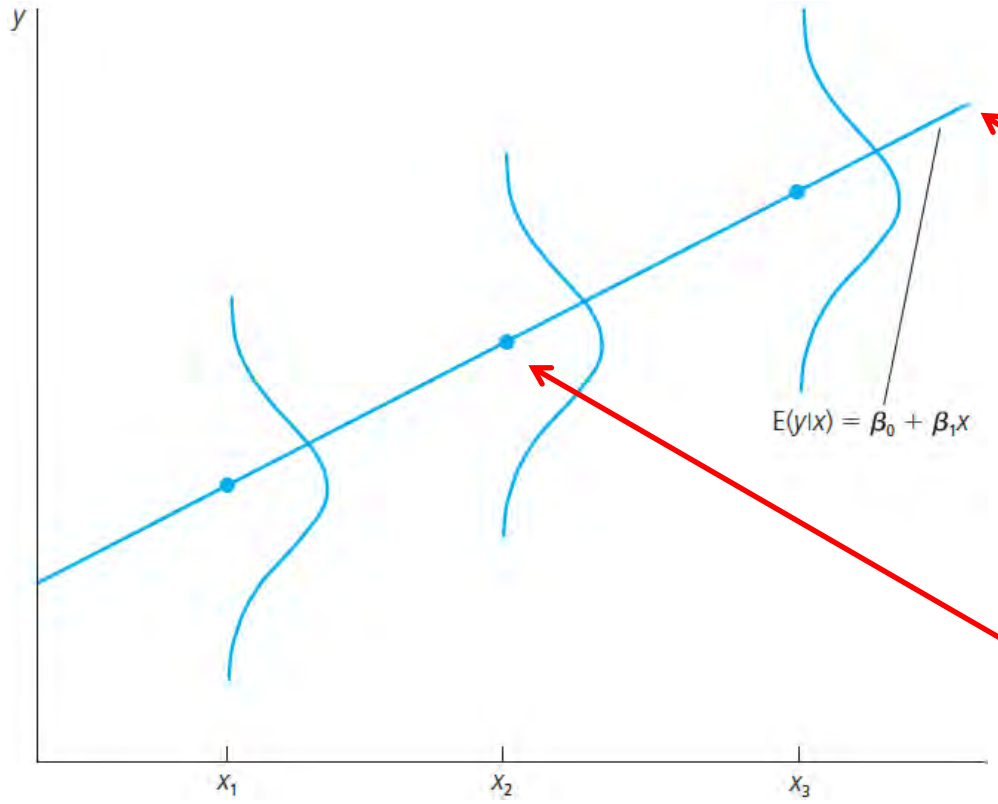
$$E(y|x) = E(\beta_0 + \beta_1 x + u|x)$$

$$= \beta_0 + \beta_1 x + E(u|x) = \beta_0 + \beta_1 x + E(u)$$

$$= \beta_0 + \beta_1 x$$

  - This means that the average value of the dependent variable can be expressed as a linear function of the explanatory variable

Population regression function (PFR)



Population regression function

For individuals with $x = x_2$, the average value of y is $\beta_0 + \beta_1 x_2$

考慮平均量

Definition of the simple linear regression model

- **Example: College versus High School GPA.**

  Suppose for the population of students attending an university, we (somehow) know

$$E(colGPA|hsGPA) = 1.5 + 0.5\ hsGPA,$$

  If $hsGPA = 3.6$ then the **average** of $colGPA$ among students with this particular high school GPA is

$$1.5 + 0.5(3.6) = 3.3$$

- Regression analysis is essentially about explaining effects of explanatory variables on **average** outcomes of y .

Deriving the Ordinary Least Squares Estimate

- **Deriving the Ordinary Least Squares Estimate**

  In order to estimate the regression model one needs data

- A random sample of n observations

(x_1, y_1) ← First observation

(x_2, y_2) ← Second observation

⋮

(x_n, y_n) ← n-th observation

$\{(x_i, y_i) : \ i = 1, \ldots n\}$

Value of the explanatory variable of the i-th observation

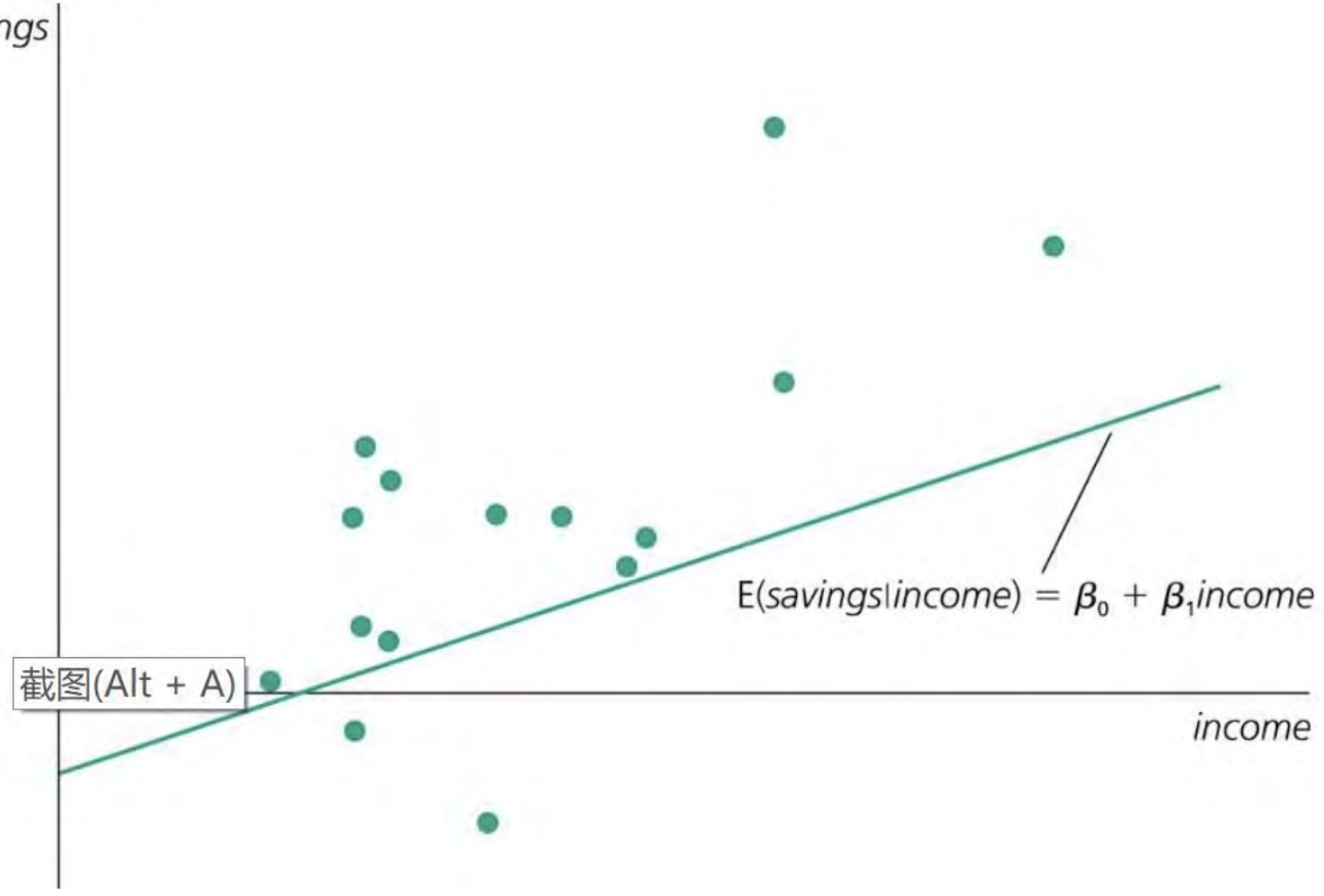
Value of the dependent variable of the i-th observation

- Using a random sample, we can generate estmators $\hat{\beta}_0$ and $\hat{\beta}_1$ to estimate the population parameters β_0 and β_1.
- We put a "hat" on the beta to denote an estimator

Deriving the Ordinary Least Squares Estimate

- **Example: Saving and income**

  This graph shows n = 15 families and the population regression of saving on income.



Deriving the Ordinary Least Squares Estimate

- **Deriving the OLS estimate**
- Where does the name "**Ordinary Least Squares**" come from?

  For any candidates $\hat{\beta}_0$ and $\hat{\beta}_1$, define a **fitted (or predicted) value** for each data point i as:

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

  We have n of these. It is the value we predict for y_i given that x has taken on the value x_i.

- The mistake we make is the residual:

$$\hat{u}_i = y_i - \hat{y}_i = y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i,$$

  and we have n residuals.

Deriving the Ordinary Least Squares Estimate

- **What does "as good as possible" mean?**
- Regression residuals

$$\hat{u}_i = y_i - \hat{y}_i = y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i$$

- Choose $\hat{\beta}_0$ and $\hat{\beta}_1$ that minimize sum of squared regression residuals

$$\min \sum_{i=1}^{n} \hat{u}_i^2 \quad \rightarrow \min \sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

- First order conditions

$$n^{-1} \sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0 \quad \leftarrow \hat{\beta}_0$$

$$n^{-1} \sum_{i=1}^{n} x_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0 \quad \leftarrow \hat{\beta}_1$$

Deriving the Ordinary Least Squares Estimate

- **First equation**

$$
\begin{aligned}
n^{-1}\sum_{i=1}^{n}(y_i-\hat{\beta}_0-\hat{\beta}_1x_i) &= n^{-1}\sum_{i=1}^{n}y_i-n^{-1}\sum_{i=1}^{n}\hat{\beta}_0-n^{-1}\sum_{i=1}^{n}\hat{\beta}_1x_i \\
&= n^{-1}\sum_{i=1}^{n}y_i-\hat{\beta}_0-\hat{\beta}_1\left(n^{-1}\sum_{i=1}^{n}x_i\right) \\
&= \bar{y}-\hat{\beta}_0-\hat{\beta}_1\bar{x}=0
\end{aligned}
$$

implies

$$\bar{y}=\hat{\beta}_0+\hat{\beta}_1\bar{x}$$

and plug this into the **second equation**

$$\sum_{i=1}^{n}x_i[y_i-(\bar{y}-\hat{\beta}_1\bar{x})-\hat{\beta}_1x_i]=0$$

Deriving the Ordinary Least Squares Estimate

Simple algebra gives

$$\sum_{i=1}^{n} x_i(y_i - \bar{y}) = \hat{\beta}_1 \left[\sum_{i=1}^{n} x_i(x_i - \bar{x}) \right] \qquad (1)$$

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n} x_i(y_i - \bar{y})}{\sum_{i=1}^{n} x_i(x_i - \bar{x})}$$

Because

$$\sum_{i=1}^{n} (x_i - \bar{x}) = 0$$

$$\sum_{i=1}^{n} x_i(y_i - \bar{y}) = \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y}) = \sum_{i=1}^{n} (x_i - \bar{x}) y_i$$

$$\sum_{i=1}^{n} x_i(x_i - \bar{x}) = \sum_{i=1}^{n} (x_i - \bar{x})^2$$

one can use equation (1) to derive:

- **Ordinary Least Squares (OLS) estimates**

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n} (x_i - \bar{x})^2}, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

Deriving the Ordinary Least Squares Estimate

- Once we have the numbers $\hat{\beta}_0$ and $\hat{\beta}_1$ for a given data set, we write the **OLS regression line** as a function of x:

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

- The OLS regression line allows us to predict y for any (sensible) value of x. It is also called the **sample regression function**.

  - The intercept, $\hat{\beta}_0$, is the predicted y when $x = 0$.

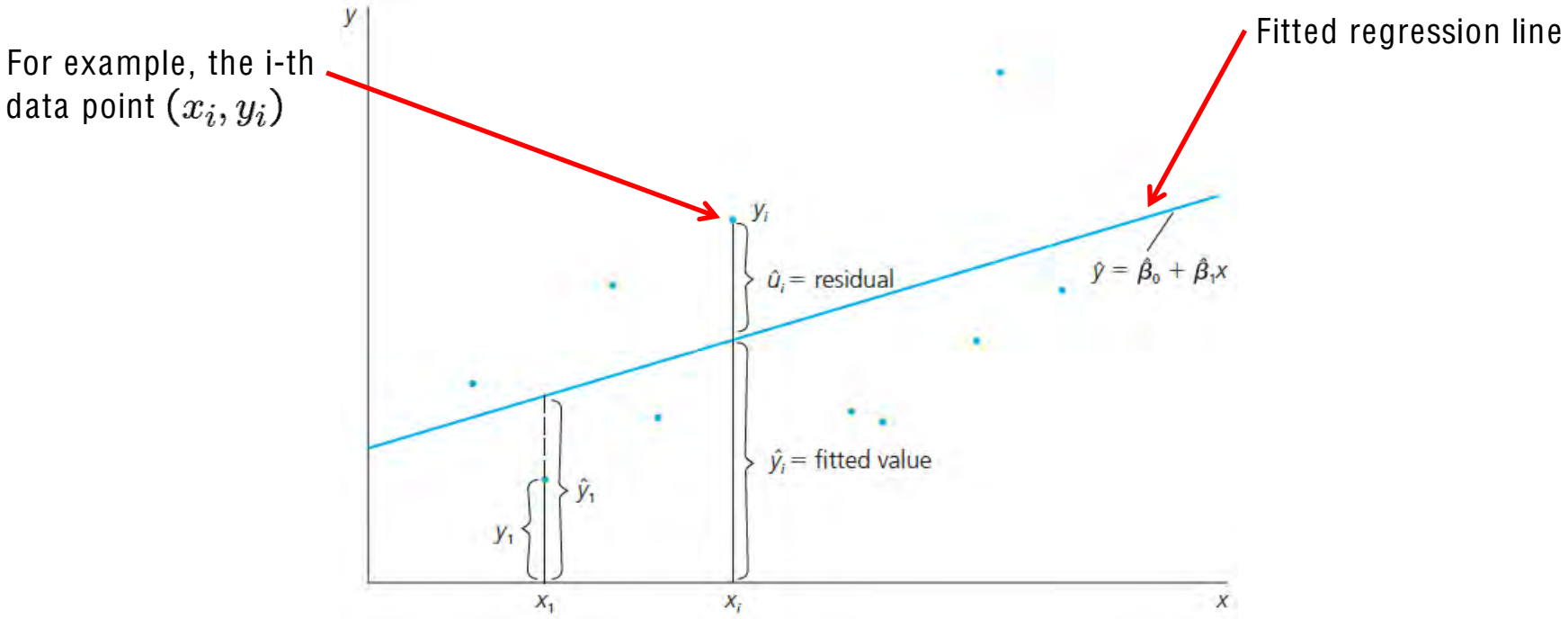
  - The slope, $\hat{\beta}_1$, allows us to predict changes in y for any (reasonable) change in x, so

$$\Delta\hat{y} = \hat{\beta}_1 \Delta x$$

If $\Delta x = 1$, so that x increases by one unit, then $\Delta\hat{y} = \hat{\beta}_1$.

Deriving the Ordinary Least Squares Estimate

- Fit as good as possible a regression line through the data points:



Deriving the Ordinary Least Squares Estimate

- CEO Salary and return on equity

$$salary = \beta_0 + \beta_1 roe + u$$

Salary in thousands of dollars

Average return on equity of the CEO's firm

- Fitted regression

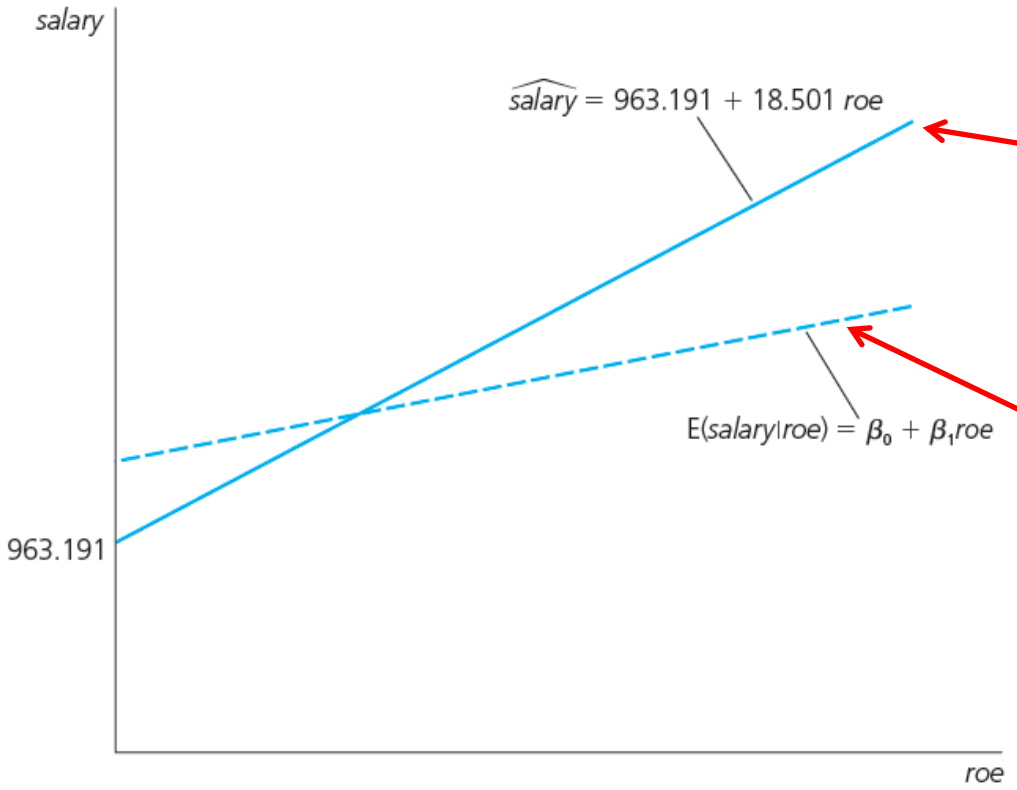
$$\widehat{salary} = 963.191 + 18.501\ roe$$

Intercept

If the return on equity increases by 1 percent, then salary is predicted to change by $18,501

- Causal interpretation?

Deriving the Ordinary Least Squares Estimate



Fitted regression line
(depends on sample)

Unknown population regression line

Deriving the Ordinary Least Squares Estimate

- Wage and education

$$wage = \beta_0 + \beta_1 educ + u$$

Hourly wage in dollars

Years of education

- Fitted regression

$$\widehat{wage} = -0.90 + 0.54\ educ$$

Intercept

In the sample, one more year of education was associated with an increase in hourly wage by $0.54

- Causal interpretation?

Deriving the Ordinary Least Squares Estimate

- Voting outcomes and campaign expenditures (two parties)

$$voteA = \beta_0 + \beta_1 shareA + u$$

Percentage of vote for candidate A

Percentage of campaign expenditures candidate A

- Fitted regression

$$\widehat{voteA} = 26.81 + 0.464\ shareA$$

Intercept

If candidate A's share of spending increases by one percentage point, he or she receives 0.464 percentage points more of the total vote

- Causal interpretation?

Properties of OLS on any sample of data

- Properties of OLS on any sample of data
- Fitted values and residuals

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i \qquad \hat{u}_i = y_i - \hat{y}_i$$

Fitted or predicted values

Deviations from regression line (= residuals)

- Algebraic properties of OLS regression

$$\sum_{i=1}^{n} \hat{u}_i = 0$$

Deviations from regression line sum up to zero

$$\sum_{i=1}^{n} x_i \hat{u}_i = 0$$

Covariance between deviations and regressors is zero

$$\bar{y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{x}$$

Sample averages of y and x lie on regression line

Properties of OLS on any sample of data

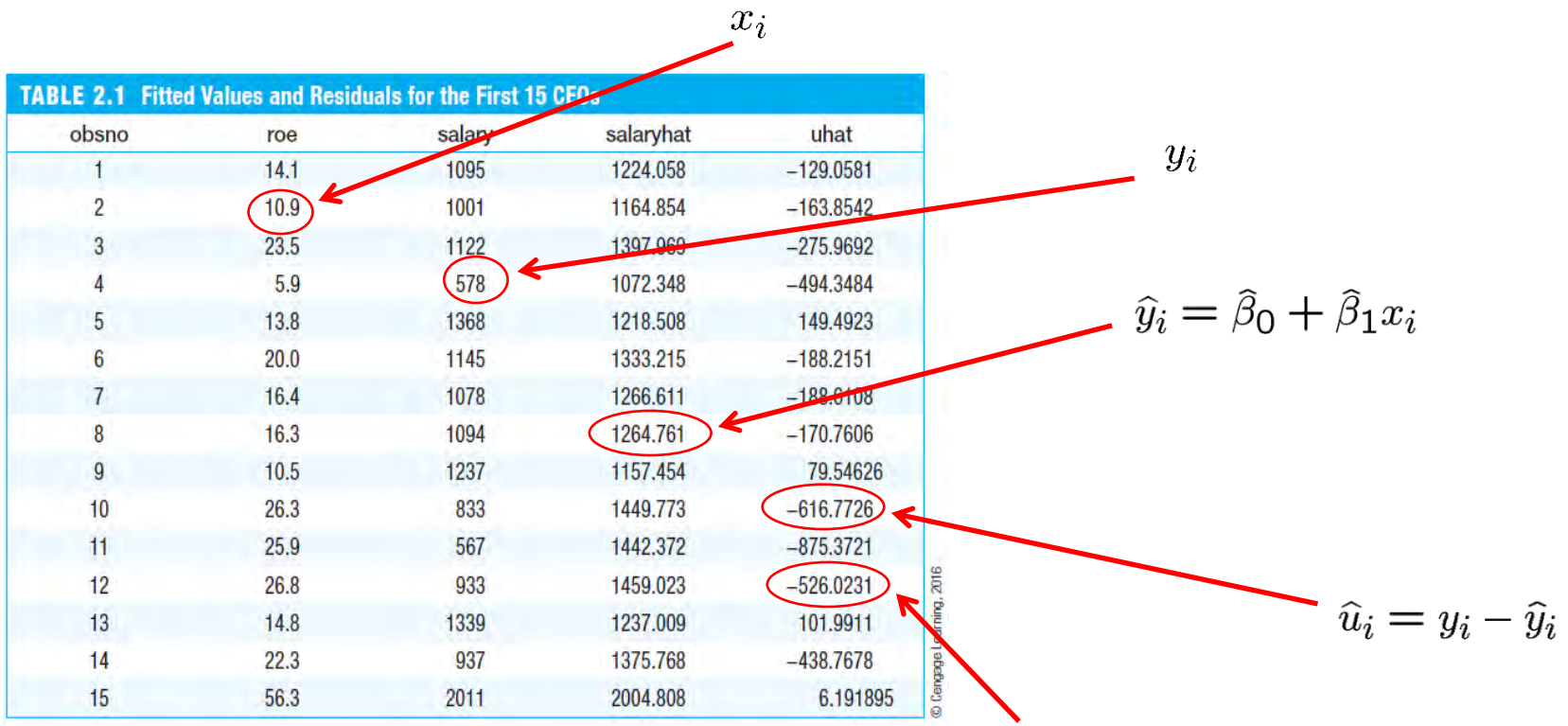


TABLE 2.1 Fitted Values and Residuals for the First 15 CEOs

obsno	roe	salary	salaryhat	uhat
1	14.1	1095	1224.058	−129.0581
2	10.9	1001	1164.854	−163.8542
3	23.5	1122	1397.969	−275.9692
4	5.9	578	1072.348	−494.3484
5	13.8	1368	1218.508	149.4923
6	20.0	1145	1333.215	−188.2151
7	16.4	1078	1266.611	−188.6108
8	16.3	1094	1264.761	−170.7606
9	10.5	1237	1157.454	79.54626
10	26.3	833	1449.773	−616.7726
11	25.9	567	1442.372	−875.3721
12	26.8	933	1459.023	−526.0231
13	14.8	1339	1237.009	101.9911
14	22.3	937	1375.768	−438.7678
15	56.3	2011	2004.808	6.191895

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x_i

y_i

$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$

$\hat{u}_i = y_i - \hat{y}_i$

For example, CEO number 12‘s salary was $526,023 lower than predicted using the the information on his firm‘s return on equity

Properties of OLS on any sample of data

- Goodness-of-Fit

"How well does the explanatory variable explain the dependent variable?"

$$y_i = \hat{y}_i + \hat{u}_i$$

- Measures of Variation

$$SST \equiv \sum_{i=1}^{n} (y_i - \bar{y})^2$$

Total sum of squares, represents total variation in the dependent variable

$$SSE \equiv \sum_{i=1}^{n} (\hat{y}_i - \bar{y})^2$$

Explained sum of squares, represents variation explained by regression

$$SSR \equiv \sum_{i=1}^{n} \hat{u}_i^{\,2}$$

Residual sum of squares, represents variation not explained by regression

Properties of OLS on any sample of data

- **Decomposition of total variation**

$$\begin{aligned} SST &= \sum_{i=1}^{n}(y_i - \bar{y})^2 = \sum_{i=1}^{n}[(y_i - \hat{y}_i) + (\hat{y}_i - \bar{y})]^2 \\ &= \sum_{i=1}^{n}[\hat{u}_i + (\hat{y}_i - \bar{y})]^2 \end{aligned}$$

using the fitted values and residuals are uncorrelated, can show:

$$SST = SSE + SSR$$

Total variation — Explained part — Unexplained part

Properties of OLS on any sample of data

- **Goodness-of-fit measure (R-squared)**

$$R^2 = \frac{SSE}{SST} = 1 - \frac{SSR}{SST}$$

R-squared measures the fraction of the total variation that is explained by the regression

$$0 \leq R^2 \leq 1$$

- $R^2 = 0$ means no linear relationship between y_i and x_i. $R^2 = 1$ means a perfect linear relationship
- As R^2 increases, the y_i are closer and closer to falling on the OLS regression line
- R^2 is a useful summary measure but tells us nothing about causality. Having a high R^2 is neither necessary nor sufficient to infer causality.

Properties of OLS on any sample of data

- CEO Salary and return on equity

$$\widehat{salary} = 963.191 + 18.501\ roe$$

$$n = 209, \quad R^2 = 0.0132$$

The regression explains only 1.3% of the total variation in salaries

- Voting outcomes and campaign expenditures

$$\widehat{voteA} = 26.81 + 0.464\ shareA$$

$$n = 173, \quad R^2 = 0.856$$

The regression explains 85.6% of the total variation in election outcomes

- Caution: A high R-squared does not necessarily mean that the regression has a causal interpretation!

Units of Measurement and Functional Form

- **Units of Measurement**

  It is very important to know how y and x are measured in order to interpret regression functions.

  Example: CEO Salary and return on equity

  $$\widehat{salary} = 963.191 + 18.501\ roe$$

  Salary in thousands of dollars

  Return on equity (In percent) of the CEO's firm

- What if we measure roe as a decimal, rather than a percent? Define

  $$roedec = roe/100$$

  Does it change the regression results?

Units of Measurement and Functional Form

- **The original regression**

$$\begin{aligned} \widehat{salary} &= 963.191 + 18.501\, roe \\ n &= 209, \quad R^2 = 0.0132 \end{aligned}$$

- **The new regression**

$$\begin{aligned} \widehat{salary} &= 963.191 + 1,850.1\, roedec \\ n &= 209, \quad R^2 = 0.0132 \end{aligned}$$

Now one percentage point change in roe is the same as 0.01 percentage points change in $roedec$ so we get the same effect as before.

Units of Measurement and Functional Form

- **Changing units of measurement for dependent variable**

  What if we measure salary in dollars, rather than thousands of dollars

$$salarydol = 1,000 \cdot salary$$

The new regression:

$$\begin{aligned} \widehat{salarydol} &= 963,191 + 18,501\ roe \\ n &= 209,\ R^2 = .0132 \end{aligned}$$

Changing units of measurement for both dependent and independent variable?

Units of Measurement and Functional Form

- **Using the Natural Logarithm in Simple Regression**

  Recall the wage example:

$$\widehat{wage} = -0.90 + 0.54\ educ$$
$$n = 526,\ R^2 = 0.1648$$

- Problem of this linear model: the dollar value of another year of schooling is constant. We expect additional years of schooling to be worth more, in dollar terms, than previous years.

- **How can we incorporate an increasing effect?**

  One way is to postulation a constant **percentage** effect. We can approximate percentage changes using the natural log.

Units of Measurement and Functional Form

- **Regression of log wages on years of eduction**

$$\log(wage) = \beta_0 + \beta_1 educ + u$$

Natural logarithm of wage

- This changes the interpretation of the regression coefficient:
  Holding u fixed,

$$\Delta \log(wage) = \beta_1 \Delta educ$$

Useful result from calculus

$$100 \cdot \Delta \log(wage) \approx \%\Delta wage$$

Percentage change of wage

therefore

$$\beta_1 = \frac{\Delta \log(wage)}{\Delta educ} \approx \frac{100^{-1} \cdot \%\Delta wage}{\Delta educ}$$

Units of Measurement and Functional Form

- **Using the Natural Logarithm in Simple Regression (cont.)**

$$100\beta_1 \approx \%\Delta wage \text{ when } \Delta educ = 1$$

In this example, $100\beta_1$ is often called the return to education. This measure is free of units of measurements.

- **The OLS regression**

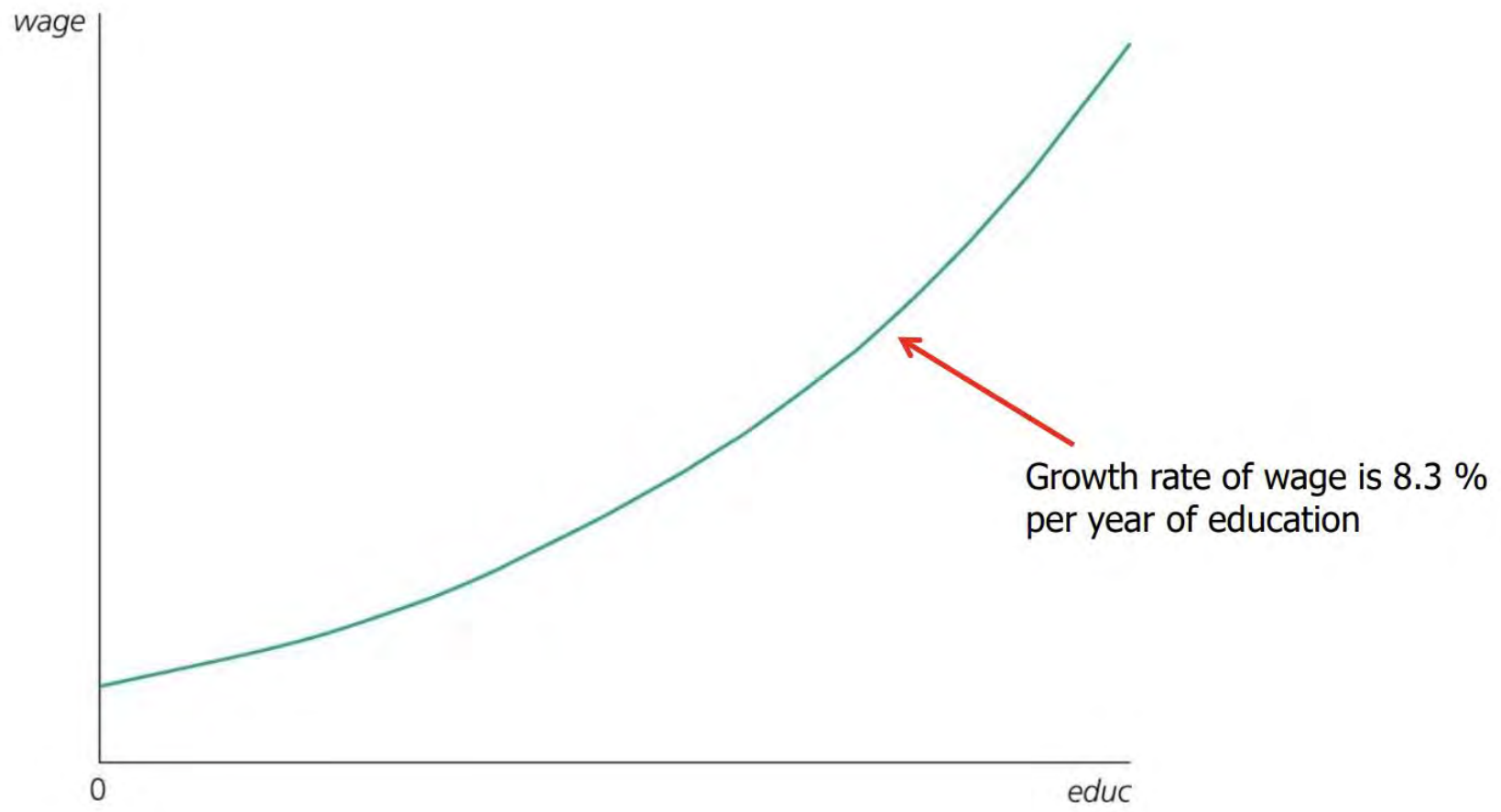
$$\widehat{\log}(wage) = 0.584 + 0.083\ educ$$

$$n = 526, \quad R^2 = 0.1858$$

The wage increases by 8.3 % for every additional year of education (= return to education)

Units of Measurement and Functional Form

- **The relationship between wage and education**



Units of Measurement and Functional Form

- **Constant elasticity model**

  For example, CEO salary and firm sales.

$$\log(salary) = \beta_0 + \beta_1 \log(sales) + u$$

  Then

$$\beta_1 \approx \frac{\%\Delta salary}{\%\Delta sales}$$

- **The OLS regression**

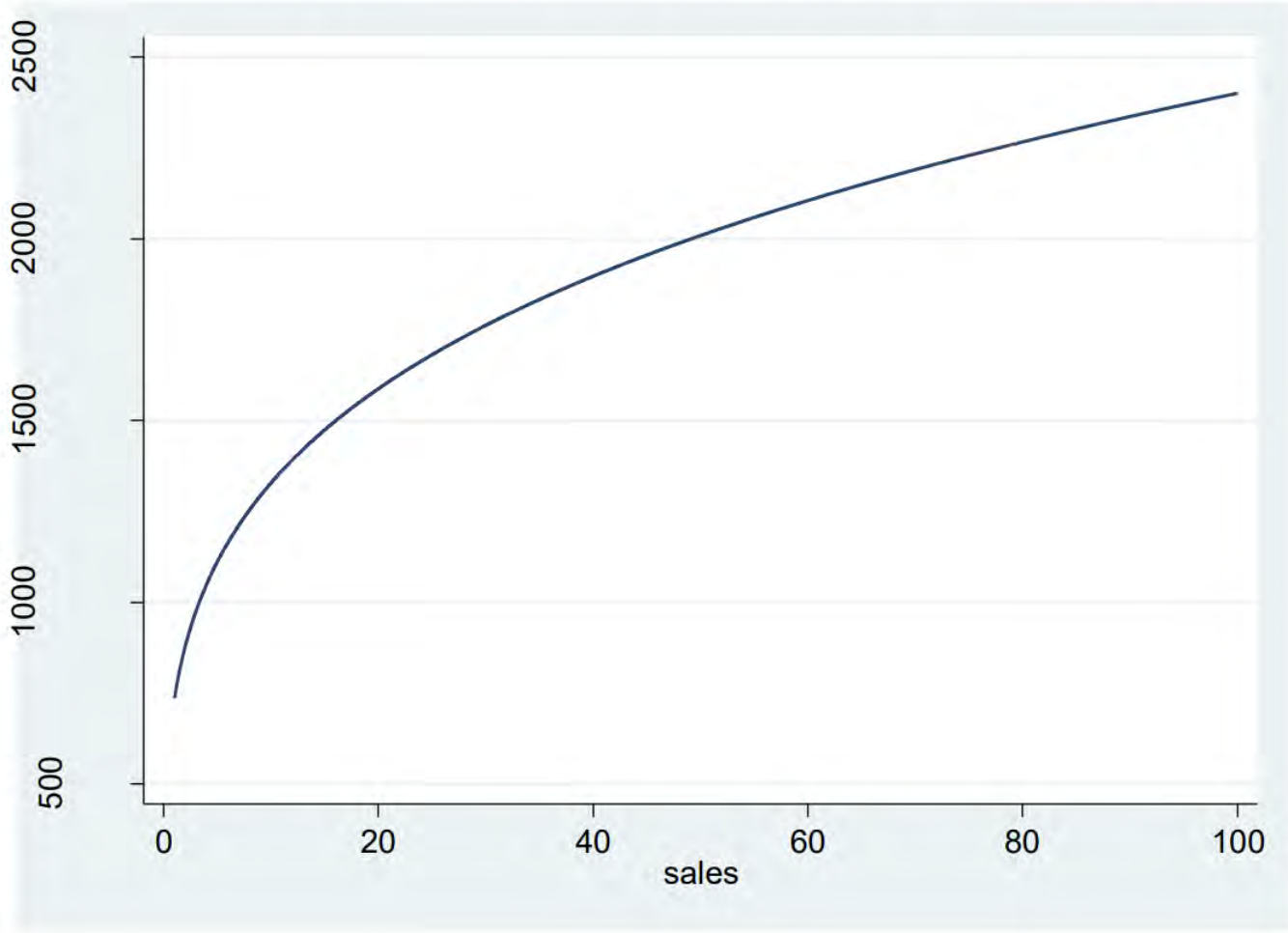
  Data CEOSAL1.dta

$$\widehat{\log}(salary) = 4.822 + 0.257 \log(sales)$$

The salary increases by 0.257 % for every 1% increase in sales

Units of Measurement and Functional Form

- **The relationship between CEO salary and firm sales.**



Units of Measurement and Functional Form

- **Summay of Functional Forms involving Logarithms**

Model	Dep. Var.	Indep. Var.	β_1
Level-Level	y	x	$\Delta y = \beta_1 \Delta x$
Level-Log	y	$\log(x)$	$\Delta y = (\beta_1/100)\%\Delta x$
Log-Level	$\log(y)$	x	$\%\Delta y = (100\beta_1)\Delta x$
Log-Log	$\log(y)$	$\log(x)$	$\%\Delta y = \beta_1 \%\Delta x$